SIMATS ENGINEERING
ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5104

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 2 **Total Marks:100**

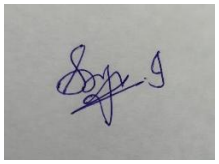
Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I **Sowmiyan I (192373014)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
Deduce the most likely original message.
Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
Demonstrate how the encryption process transforms the message step-by-step.
Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
Analyze how investigators could detect the presence of hidden data within the image.
Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

SCENARIO 1 – EVENT COMMUNICATION SECURITY

Scenario: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

1. Understanding of the Concept

The classical encryption technique applicable to this scenario is the Caesar cipher. It is a substitution cipher in which every alphabetic character is replaced by another character located a fixed number of positions away in the alphabet. In this problem, the fixed shift is 4 positions. Therefore, each plaintext letter is moved four positions forward to generate the ciphertext.

For example, A becomes E, B becomes F, C becomes G, and so on. When the end of the alphabet is reached, the process wraps around to the beginning. Thus, X becomes B, Y becomes C, and Z becomes D when the shift is 4. Spaces between words are not encrypted and remain unchanged.

The Caesar cipher is useful for demonstrating the fundamental idea of substitution-based encryption, although it is not suitable for protecting highly confidential information in modern communication systems because its possible key space is very small.

2. Encryption Rule

Let each alphabetic character be represented by a number from 0 to 25, where A = 0, B = 1, ..., Z = 25. For a shift value of 4, the encryption formula is:

C = (P + 4) mod 26

Here, P represents the numerical value of the plaintext character and C represents the numerical value of the encrypted character. The modulo operation ensures that the alphabet wraps around after Z.

3. Step-by-Step Encryption of “MEET AT NOON”

Plaintext	Numerical Value	Shift	Ciphertext	Movement
M	12	+4	Q	M → N → O → P → Q
E	4	+4	I	E → F → G → H → I
E	4	+4	I	E → F → G → H → I
T	19	+4	X	T → U → V → W → X
A	0	+4	E	A → B → C → D → E
T	19	+4	X	T → U → V → W → X
N	13	+4	R	N → O → P → Q → R
O	14	+4	S	O → P → Q → R → S
O	14	+4	S	O → P → Q → R → S
N	13	+4	R	N → O → P → Q → R

Final ciphertext: QI IX SRRS

Therefore, after applying a shift of 4 to every alphabetic character and preserving the spaces, the encrypted form of “MEET AT NOON” is “QI IX SRRS”.

4. Decryption by an Authorized Member

An authorized member who knows the encryption rule and the shift value can recover the original message. Since encryption moved each character four positions forward, decryption moves every character four positions backward.

P = (C - 4) mod 26

For example, Q is shifted four positions backward: Q → P → O → N → M, so Q becomes M.

Similarly, I becomes E, X becomes T, R becomes N, and S becomes O.
Applying the reverse operation to the complete ciphertext gives:

QI IX SRRS → MEET AT NOON

5. Experimental Design / Implementation

To test the encryption system, first select the plaintext message and define the fixed key as 4. Next, convert each alphabetic character into its numerical position, apply the Caesar formula, and convert the resulting value back into a letter. Spaces should be copied without modification. The ciphertext should then be passed to a separate decryption step using the reverse formula.
A good validation procedure is to perform a round-trip test: plaintext → encryption → ciphertext → decryption → recovered plaintext. If the final recovered plaintext exactly matches the original message, the implementation is functioning correctly for the tested input.

6. Example Python Implementation

```
def caesar_encrypt(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord("A") if ch.isupper() else ord("a")
            result += chr((ord(ch) - base + shift) % 26 + base)
        else:
            result += ch
    return result

message = "MEET AT NOON"
encrypted = caesar_encrypt(message, 4)
print(encrypted) # QI IX SRRS
```

7. Tools and Technologies

The experiment can be implemented using Python in an IDE such as Visual Studio Code, PyCharm, or any Python environment. The implementation mainly requires basic string processing and character conversion functions. No external cryptographic package is necessary for demonstrating this classical cipher. A text editor can also be used for manual verification.

8. Analysis of Results

The plaintext “MEET AT NOON” is successfully transformed into “QI IX SRRS” using a fixed shift of 4. Applying the reverse shift restores the exact original message. This confirms that the same known shift value can be used as the basis for reversing the transformation.
The result also demonstrates the weakness of the Caesar cipher. Since the alphabet contains only 26 positions, an attacker can try all possible shifts and identify the meaningful plaintext. Frequency and language patterns can also assist an attacker. Therefore, while the method is useful for learning the principle of substitution encryption, it does not provide strong confidentiality for modern sensitive communication.

9. Conclusion

The event message can be encrypted using a Caesar cipher with a shift of 4, producing “QI IX SRRS”. An authorized member who knows the shift can decrypt it by moving each letter four positions backward, recovering “MEET AT NOON”. The experiment demonstrates encryption, decryption, validation, and the security limitations of a simple classical cipher.

SCENARIO 2 – INTERCEPTED SUSPICIOUS MESSAGE

Scenario: A cybersecurity intern intercepts the coded message “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption. Deduce the most likely original message. Justify the answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

1. Understanding of the Concept

The intercepted message is most consistent with the Atbash cipher, a simple classical substitution technique. Atbash replaces each letter with its corresponding letter from the reversed alphabet. The mapping begins with A ↔ Z, B ↔ Y, C ↔ X, D ↔ W and continues until M ↔ N.
Unlike a Caesar cipher, Atbash does not use a numerical shift. Instead, it uses a fixed reversed

alphabet. Because the substitution relationship is symmetrical, applying the same mapping again decrypts the ciphertext.

2. Atbash Mapping

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

Thus, for example, G maps to T, S maps to H, R maps to I, and H maps to S. Applying these substitutions to each word reveals a meaningful English sentence.

3. Step-by-Step Decryption

Ciphertext Word	Character Mapping	Plaintext Word	Reason
GSRH	G→T, S→H, R→I, H→S	THIS	Valid English word
RH	R→I, H→S	IS	Common 2-letter word
Z	Z→A	A	Valid 1-letter word
HVXIVG	H→S, V→E, X→C, I→R, V→E, G→T	SECRET	Valid 6-letter word

Most likely original message: “THIS IS A SECRET”

4. Pattern and Letter-Structure Analysis

The first important clue is the word-length pattern: 4–2–1–6. After applying the Atbash mapping, the same pattern becomes “THIS IS A SECRET”, which forms a grammatically meaningful English sentence.

The repeated two-letter ciphertext word “RH” is particularly useful. Under the Atbash mapping, R becomes I and H becomes S, producing “IS”. The single-letter ciphertext “Z” becomes “A”, which is one of the common one-letter English words. The four-letter ciphertext “GSRH” becomes “THIS”, and the six-letter word “HVXIVG” becomes “SECRET”.

The consistency of all four word transformations provides strong internal evidence for the Atbash interpretation. The resulting sentence is meaningful without requiring any changes to the spaces or word boundaries.

5. Experimental Design / Verification

To experimentally verify the result, construct the reversed alphabet mapping and apply it to every alphabetic character in the ciphertext. Preserve spaces and word boundaries. After obtaining the candidate plaintext, re-apply the same Atbash mapping to the plaintext. If the original ciphertext is reconstructed exactly, the substitution process has been verified.

A second validation method is structural checking. Compare the word lengths before and after decryption and check whether the resulting sequence contains valid words and a meaningful sentence. For this message, both tests succeed.

6. Example Python Implementation

```
def atbash_decrypt(text):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord("A") if ch.isupper() else ord("a")
            result += chr(base + 25 - (ord(ch) - base))
        else:
            result += ch
    return result

ciphertext = "GSRH RH Z HVXIVG"
print(atbash_decrypt(ciphertext))  # THIS IS A SECRET
```

7. Tools and Technologies

The investigation can be performed manually using an Atbash alphabet table or programmatically using Python. Python provides character-processing functions that make it possible to automate the substitution and verify the result. A simple IDE or terminal is sufficient for execution.

8. Analysis of Results

The ciphertext “GSRH RH Z HVXIVG” decrypts consistently to “THIS IS A SECRET”. Every letter follows the reversed-alphabet substitution rule, and the word boundaries produce a meaningful sentence. The combination of character mapping, word lengths, repeated patterns, and semantic consistency supports the identification of Atbash as the most likely encryption mechanism.

From a security perspective, the message is not strongly protected because the substitution rule is fixed and easy to reproduce. Once the analyst recognizes the Atbash relationship, the message can be decrypted without needing to search through a large key space. This demonstrates why simple classical substitution techniques are mainly educational rather than suitable for modern confidential communication.

9. Conclusion

The most likely original message is “THIS IS A SECRET”. The Atbash cipher explains the transformation because each ciphertext character maps to the corresponding character in the reversed alphabet. The repeating patterns, word lengths, and meaningful resulting sentence provide evidence for the deduction. The experiment also demonstrates the limitations of fixed substitution techniques in protecting sensitive information.